

D-stability under first-order attachments at a single coordinate: an exact static threshold and a subset-sum criterion

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Abstract

A real matrix is D-stable if it remains Hurwitz under every positive diagonal rescaling, that is, for every choice of positive time scales of its coordinates. We study a D-stable core that interacts with its environment through a single coordinate, the port, to which a total load G of first-order attachments is coupled. Binding sites, decoys, buffers and relaxing partner species are examples. The question is whether it matters how the load is split into pieces with independent time scales. It does, and we characterize exactly how. Let τ be the onset of instability along the static loading ray, the infimum of the loads X at which $B + Xe_p e_p^T$ is strictly D-unstable. Every partition of a total $G < \tau$ is D-stable. At $G = \tau$ only a zero eigenvalue can occur. For $G > \tau$ every sufficiently fine partition has a strictly growing mode at finite, explicitly constructed rates. This threshold theorem is verified in Lean 4. For a fixed partition, a closed-right-half-plane absorption identity reduces D-stability to planar geometry: the core's port response must stay below the lower envelope of semicircles erected over the subset sums of the partition. Three consequences follow. A partition is strictly unstable whenever one of its subset sums lies in the static instability window. Merging pieces never destroys D-stability. For an explicit four-state family, the subset-sum test is exact throughout the relevant range of loads. In that family a load of 2 is safe for all time scales if and only if a single piece carries at least $(7963 + 329\sqrt{409})/7780 \approx 1.8787$ of it: the split $(19/10, 1/10)$ is D-stable, while $(15/8, 1/8)$ and $(1, 1)$ are not. Four core states are necessary for such granularity sensitivity. Bounding the ratio of time scales strictly raises the threshold when $\det(B + \tau e_p e_p^T) \neq 0$. We extend the safety results to several ports and to reciprocal relaxation modules, and we relate them to negative-imaginary systems theory.

Keywords: D-stability; diagonal scaling; Hurwitz stability; robust stability; value sets; subset sums; negative-imaginary systems; buffers and decoys; formal verification.

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1 Introduction

D-stability and unknown time scales. Let $\dot{x} = Bx$ be the linearization of a dynamical system at a steady state. In many applications the entries of B are known only up to the speed of each variable: a species' compartment volume, turnover time or buffering capacity rescales its row. Stability that survives every such choice is *D-stability*: DB is Hurwitz for every positive diagonal matrix D . The notion goes back to Enthoven and Arrow [13] and Arrow and McManus [2] in the stability of competitive equilibria. It has since become a standard object in matrix theory, ecology and biochemistry [21, 9, 4, 17, 28, 18, 25]. Characterizing D-stability by finitely many algebraic conditions is classical in dimension three [6] and known in dimension four [22, 19]. In general it remains open [25, 26]. Related robust notions include strong D-stability [1] and structural stability of biochemical networks [5]. Recent work on chemical reaction networks locates instability in small “unstable cores” [32].

The question. This paper does not attempt the general characterization. It asks a narrower question, which arises whenever a stable subsystem is loaded through one of its variables. Picture a *core* B , for example a small gene or signalling circuit, which is D-stable on its own. One of its variables, the *port* p , is shared with an environment of *attachments*. Each attachment is a single variable that is driven by the port, relaxes on its own time scale and feeds back into the port (Figure 1). A reversible binding site, a decoy, a buffer and a sequestering partner are all of this type. Every attachment carries a *load* $r_j > 0$, which is the product of its forward and return gains divided by its own damping; Section 2.3 makes this precise. The total load is $G = \sum_j r_j$. Now fix G . Is the D-stability of the whole system determined by G alone, or does it matter whether the load arrives as one lump or as many fragments with independent time scales?

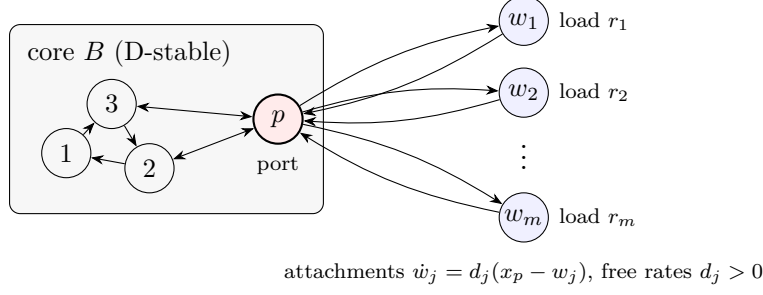


Figure 1: The interface architecture. A D-stable core interacts with m first-order attachments only through one coordinate, the port p . Attachment j contributes load r_j to the port equation. Every core and attachment variable has its own unknown positive rate. The partition $r = (r_1, \dots, r_m)$ of the total load G is the object of study.

The central mechanism. The answer turns on a simple observation. An attachment that is much faster than the port behaves like an instantaneous extra term on the port's diagonal: it shifts the port's self-coefficient by its load. An attachment that is much slower than the port is effectively frozen and contributes nothing. An attachment of intermediate speed contributes a phase lag. We show that this lag can always be absorbed into the port's own free time scale (Lemma 3.1). The dynamic problem therefore collapses onto the *static loading ray* $B_X = B + X e_p e_p^T$, $0 \leq X \leq G$. D-stability need not be monotone along this ray: B_X can be D-stable at both ends and strictly D-unstable in a middle window. A single lump cannot produce an intermediate effective load without also producing a large phase lag, and that lag protects it (Section 5). Fragments can instead be divided into a fast group and a slow group. If a *subset sum* of the pieces lies in the unstable window, making that subset fast and the rest slow reproduces an unstable static matrix (Figure 2).

Main results. Write τ for the onset of strict D-instability along the ray, the infimum (2) of the loads at which B_X is strictly D-unstable.

- (A) *Exact threshold* (Theorem 4.3, verified in Lean 4). Every positive partition of a total $G < \tau$ is D-stable. At $G = \tau$ the attached system is D-stable if and only if $\det B_\tau \neq 0$. For $G > \tau$ there is $\varepsilon > 0$ such that every partition into pieces smaller than ε has a strictly unstable positive scaling. The unstable eigenvalue of the static matrix is reproduced exactly, with rates given in closed form (Lemma 4.2).
- (B) *Fixed partitions* (Theorem 5.2, Corollary 5.3). A fixed partition is D-stable if and only if a determinant sign holds and the core's port response stays below the lower envelope of an explicit family of semicircles over the subset sums. Equivalently, the response must lie in a family of disks, a disk form reminiscent of the circle criterion. Consequences (Section 6) are a subset-sum instability criterion, an exact reduction of every verdict to one-attachment “edges” and static “corners”, and monotonicity under merging.
- (C) *An exactly solvable family* (Theorem 8.3, Corollary 8.4). For an explicit rational four-state family $B(h_0)$ the subset-sum criterion is exact, for any number of pieces and every total $G \leq h_0 - 4$. In particular, with total load 2 on the core $B(6)$, a partition is D-stable if and only if its largest piece is at least $6 - h_- = (7963 + 329\sqrt{409})/7780$. Four core states are the minimum for any granularity effect (Proposition 7.1).
- (D) *Bounded time-scale heterogeneity* (Proposition 4.5). Above τ , fine partitions are destabilized at a uniformly bounded ratio of time scales. If $\det B_\tau \neq 0$, any fixed bound on that ratio strictly raises the effective threshold, so the required ratio diverges as $G \downarrow \tau$.

- (E) *Extensions* (Section 9). An open-box criterion for several ports, safety for reciprocal relaxation modules, and a value-set criterion for a hub joining several D-stable modules. The latter gives a D-stability counterpart of the negative-imaginary stability theorem.

Relation to previous work. Value sets and zero exclusion are the basic tools of parametric robust control [3]. Our fixed-partition test is a value-set statement in which the uncertain parameters are the time scales. The absorption identity is what makes the value set of each attachment explicitly planar. The disk form recalls the circle criterion [35], although the disks here come from the partition rather than from a sector bound. Lanzon and Petersen [27, 34, 30] showed that a negative-imaginary feedback loop is stable exactly when its DC loop gain is suitably bounded. Our hub criterion (Theorem 9.3) has the same flavour when every module is negative imaginary for every scaling. The attachment theorem, however, does not assume the core to be negative imaginary: it retains the *first* static loss along the ray instead of an endpoint test. That is what allows a stable-to-unstable-to-stable pattern along the loading ray. Reciprocal relaxation modules are classical relaxation systems [33, 7]. Fast and slow attachments are singular-perturbation limits [24]. The robustness of loaded systems to uncertain load time constants was studied for power systems [29]. In biology, buffers can create or suppress calcium oscillations [14, 16]. Decoy binding sites and slow unbinding shape biomolecular clocks [23, 12]. Loading of a module by downstream binding, known as retroactivity, alters its dynamics [11, 20]. These phenomena are known, and we do not claim them. Our contribution is an exact linear mechanism and classification: which distributions of a fixed load can destabilize a fixed linearized core for *some* choice of time scales, and which cannot for *any*. We are not aware of a previous statement of the threshold theorem, the subset-sum criterion or the four-state minimum.

Organization and evidence. Section 2 fixes conventions and derives the normalized model from physical attachments. Sections 3–4 prove the threshold theorem. Sections 5–6 treat fixed partitions. Section 7 settles the minimum dimension, and Section 8 solves the four-state family. Section 9 contains the extensions, Section 10 the application to binding sites and buffers, and Section 11 the verification status of every result. Theorem 4.3 is formally verified. Lemma 8.2 is an exact computer-assisted certificate. All other proofs are conventional and are given in full.

2 Setting

2.1 Conventions

A real square matrix M is *Hurwitz* if all its eigenvalues satisfy $\operatorname{Re} \lambda < 0$. A *positive scaling* of M is DM with $D = \operatorname{diag}(d_1, \dots, d_N)$, $d_i > 0$. Since $DM = D(MD)D^{-1}$, left and right scalings have the same spectra; the Lean formalization uses right scalings. We say that M has *strict growth*, or has a *strictly unstable positive scaling*, if some DM has an eigenvalue with $\operatorname{Re} \lambda > 0$. M is *D-stable* if every DM is Hurwitz. D-stable matrices are nonsingular, and marginal cases lie between the two notions: a matrix without strict growth may still have a scaling with an imaginary-axis eigenvalue, which we call a *contact*.

It is often convenient to work with *inverse rates* $Q = D^{-1}$: λ is an eigenvalue of DM exactly when the pencil $\lambda Q - M$ is singular. Since M is real, contacts $\pm i\omega$ come in conjugate pairs. Replacing Q by ωQ moves a contact at $i\omega$, $\omega > 0$, to i . We use this frequency normalization repeatedly; it is legitimate only because *all* rates are free.

2.2 The attached system

Fix a D-stable $n \times n$ matrix B , a coordinate p with unit vector $e = e_p$, and a finite *partition* $r = (r_1, \dots, r_m)$ with every $r_j > 0$ and total $G = \sum_j r_j$. The attached matrix and the static loading ray are

$$A_r = \begin{pmatrix} B & er^T \\ \mathbf{1}e^T & -I_m \end{pmatrix}, \quad B_X = B + Xee^T \quad (X \in \mathbb{R}). \quad (1)$$

Attachment j obeys $\dot{w}_j = d_{n+j}(x_p - w_j)$, and the port equation receives $\sum_j r_j w_j$. All $n + m$ rates are free.

2.3 From physical attachments to normalized loads

A general scalar attachment has a port gain a_j (the effect of w_j on $\dot{x}_p$), a capture gain c_j (the effect of x_p on $\dot{w}_j$) and a damping $h_j > 0$:

$$\dot{w}_j = d_j(c_j x_p - h_j w_j), \quad \dot{x}_p = d_p((Bx)_p + a_j w_j).$$

Three changes of variables bring this to (1).

- (i) *Absorb the damping into the rate.* Put $d'_j = d_j h_j$. Since d_j is free, so is d'_j , and the attachment row becomes $d'_j((c_j/h_j)x_p - w_j)$.
- (ii) *Rescale the state.* Assume $c_j \neq 0$; otherwise the attachment does not couple back. Put $w_j = (c_j/h_j)\tilde{w}_j$. This is a diagonal similarity, and diagonal similarities commute with diagonal scalings, so D-stability is unaffected. The attachment row becomes $d'_j(x_p - \tilde{w}_j)$ and the port receives $r_j \tilde{w}_j$ with

$$r_j = \frac{a_j c_j}{h_j}.$$

- (iii) *Signs.* The algebra in (ii) is valid for either sign of c_j , so only the sign of the load r_j matters. Positive loads, $a_j c_j > 0$, are positive-feedback partners, and they are the subject of this paper. Negative loads are damping partners (Remark 5.4).

The dictionary for reversible binding sites and buffers is given in Section 10. For such sites the load is $r = ku/(u + \delta)$ for binding coefficient k , unbinding rate u and bound-complex degradation δ , and the site also adds a loss k to the port diagonal.

2.4 The static threshold

The static loading ray is the family of matrices obtained by making every attachment infinitely fast. Its first loss of stability is the quantity that governs everything below:

$$\tau = \inf\{X \geq 0 : B_X \text{ has a strictly unstable positive scaling}\}. \quad (2)$$

Lemma 2.1. $0 \leq \tau \leq -B_{pp}$.

Proof. A D-stable matrix has nonpositive diagonal: if $B_{pp} > 0$, scaling row p by 1 and all other rows by $\eta \rightarrow 0$ gives an eigenvalue near $B_{pp} > 0$. If $X > -B_{pp}$, the (p, p) entry of B_X is positive. Scaling row p by a large factor makes the trace positive, so some eigenvalue has positive real part. Hence every $X > -B_{pp}$ belongs to the set in (2). $\square$

The set in (2) is open, because a strict eigenvalue persists under small perturbations of X at a fixed scaling. It does not contain 0, because B is D-stable. Consequently, for every $X > \tau$ the interval (τ, X) contains a load with strict growth, and if $\tau > 0$ then B_τ has no strict growth.

3 Absorption and static safety

The following identity is the engine of the paper. It says that at any candidate eigenvalue in the closed right half-plane, a first-order attachment is indistinguishable from a static load *smaller* than its own, plus an increase of the port's inverse rate.

Lemma 3.1 (Closed-half-plane absorption). *Let $z = \sigma + i\omega \neq 0$ with $\sigma \geq 0$, and let $r, t > 0$. Put*

$$d = (1 + \sigma t)^2 + \omega^2 t^2, \quad \alpha = \frac{rt}{d}, \quad x = \frac{r(1 + 2\sigma t)}{d}. \quad (3)$$

Then $\alpha > 0$, $0 < x < r$, and

$$\frac{r}{1 + zt} + z\alpha = x. \quad (4)$$

Proof. Here $d = |1 + zt|^2 > 0$ and $r/(1 + zt) = r(1 + \bar{z}t)/d$. Adding $z\alpha = zrt/d$ gives $r(1 + \bar{z}t + zt)/d = r(1 + 2\sigma t)/d = x$. Finally $r - x = r(d - 1 - 2\sigma t)/d = r|z|^2 t^2/d > 0$. $\square$

Write the inverse rates of the attached system as $\hat{Q} = \text{diag}(Q, t_1, \dots, t_m)$, with $Q > 0$ the core part. If $(\lambda\hat{Q} - A_r)(v, u) = 0$ and $1 + \lambda t_j \neq 0$, the attachment rows give $u_j = v_p/(1 + \lambda t_j)$. The core rows then give

$$(\lambda Q - B - L_r(\lambda)ee^T)v = 0, \quad L_r(\lambda) = \sum_{j=1}^m \frac{r_j}{1 + \lambda t_j}. \quad (5)$$

Conversely, a nonzero solution v of (5) extends to an eigenvector. Moreover, $v = 0$ forces $u = 0$, and

$$\det(\lambda\hat{Q} - A_r) = \prod_j (1 + \lambda t_j) \cdot \det(\lambda Q - B - L_r(\lambda)ee^T).$$

At $\lambda = 0$, block elimination gives $\det(-A_r) = \det(-B_G)$.

Theorem 3.2 (Static-interval safety). *Suppose that B_X is D-stable for every $0 < X < G$. Then for every positive partition r of G and every positive scaling, every nonzero eigenvalue of the scaled A_r has negative real part. Moreover A_r is D-stable if and only if $\det B_G \neq 0$. If $\det B_G = 0$, then 0 is an eigenvalue of every positive scaling of A_r .*

Proof. Let $\lambda \neq 0$ with $\text{Re } \lambda \geq 0$ be an eigenvalue at inverse rates $\hat{Q}$. Then $1 + \lambda t_j \neq 0$, so (5) holds with $v \neq 0$. Apply Lemma 3.1 to every term of $L_r(\lambda)$: $L_r(\lambda) = X - \lambda \sum_j \alpha_j$ with $X = \sum_j x_j \in (0, G)$ and $\alpha_j > 0$. Then (5) reads $(\lambda Q' - B_X)v = 0$ with $Q' = Q + (\sum_j \alpha_j)ee^T$, again positive diagonal. So λ is a closed-right-half-plane eigenvalue of a positive scaling of B_X , contradicting D-stability. Zero is an eigenvalue if and only if $\det(-A_r) = \det(-B_G)$ vanishes, and this does not depend on the rates. $\square$

The proof treats real and complex candidates alike. It needs neither continuation along paths of rates nor any stability assumption on the principal submatrix obtained by deleting the port. At $\sigma = 0$, with the frequency normalized to $\omega = 1$, it specializes to the familiar phase absorption $X = \sum_j r_j/(1 + t_j^2)$, $q'_p = q_p + \sum_j r_j t_j/(1 + t_j^2)$.

4 The exact threshold

Lemma 4.1 (Static marginality below first strict growth). *B_X is D-stable for every $0 \leq X < \tau$.*

Proof. For $X < \tau$ there is no strict growth by definition, and $X = 0$ is the hypothesis. Suppose some $0 < X < \tau$ has a contact: $\det(i\omega Q - B_X) = 0$ for some $Q > 0$ and $\omega \in \mathbb{R}$. Put $P(z) = zQ - B$ and let $c(z) = \text{adj}(P(z))_{pp}$ be the (p, p) cofactor. The cofactor does not depend on the (p, p) entry of P . Expansion along row p gives

$$\det(P(z) - Yee^T) = \det P(z) - Y c(z) \quad (Y \in \mathbb{C}).$$

At the contact, $\det P(i\omega) \neq 0$ because B is D-stable, hence $c(i\omega) \neq 0$. The quotient $f = \det P/c$ is therefore analytic near $i\omega$, with $f(i\omega) = X$. Changing the port inverse rate from q_p to $q_p + k$ replaces $P(z)$ by $P(z) + zk ee^T$ without changing c . So the loaded, re-rated pencil is singular at z exactly when

$$f(z) + zk - Y = 0. \quad (6)$$

Take $z = \sigma + i\omega$ with small $\sigma > 0$. If $\omega \neq 0$, solve (6) by $k(\sigma) = -\text{Im } f(z)/\omega$ and $Y(\sigma) = \text{Re } f(z) + \sigma k(\sigma)$. If $\omega = 0$, take $k = 0$ and $Y = f(\sigma)$, which is real. In both cases $k(\sigma) \rightarrow 0$ and $Y(\sigma) \rightarrow X$ as $\sigma \downarrow 0$, since $f(i\omega) = X$ is real. For small σ the inverse rate $q_p + k(\sigma)$ is positive and $Y(\sigma) > 0$, and $B_{Y(\sigma)}$ has the eigenvalue z with $\text{Re } z = \sigma > 0$. Thus $\tau \leq Y(\sigma)$ for all small σ , so $\tau \leq X$, a contradiction. $\square$

The next lemma turns static growth into dynamic growth of every sufficiently fine partition, preserving the unstable eigenvalue exactly.

Lemma 4.2 (Finite realization of static growth). *Let $0 < X < G$, and suppose that B_X has an eigenvalue $z = \sigma + i\omega$ with $\sigma > 0$ at inverse rates $Q > 0$. Put $q = Q_{pp}$, $k = |z|^2$ and $f(t) = (1 + 2\sigma t)/(1 + 2\sigma t + kt^2)$. Choose $\delta > 0$ with*

$$G\delta + \frac{Gk\delta^2}{2\sigma} \leq \frac{q}{2}, \quad Gk\delta^2 < G - X, \quad (7)$$

and put $\varepsilon = \min\{G - X, X(1 - f(\delta))/2\} > 0$. Then:

- (a) every partition of G with $\max_j r_j < \varepsilon$ has a positive scaling D such that z itself is an eigenvalue of DA_r ;
- (b) the rates of (a) can be chosen in a compact subset of $(0, \infty)$ that depends on B, X, z, Q, G but not on the partition.

Proof. The function f decreases from 1 to 0 on $(0, \infty)$, since $f'(t) = -2kt(1 + \sigma t)/(1 + 2\sigma t + kt^2)^2$. Also $1 - f(\delta) = k\delta^2/(1 + 2\sigma\delta + k\delta^2) \in (0, k\delta^2]$.

Order the pieces arbitrarily and let S be the first partial sum exceeding X . Consecutive partial sums differ by less than ε , so $X < S < X + \varepsilon \leq G$; in particular some pieces remain after this group. Give the pieces of this first group inverse rate δ (“fast”), and give the remaining pieces a common inverse rate T (“slow”) determined by

$$f(T) = y := \frac{X - Sf(\delta)}{G - S}. \quad (8)$$

We check that $y \in (0, 1)$ with bounds independent of the partition. First, $Sf(\delta) < (X + \varepsilon)f(\delta) \leq Xf(\delta) + \varepsilon$, so $X - Sf(\delta) > X(1 - f(\delta)) - \varepsilon \geq X(1 - f(\delta))/2$, and $y \geq y_- := X(1 - f(\delta))/(2G) > 0$. Second, $G - S - (X - Sf(\delta)) = G - X - S(1 - f(\delta)) \geq G - X - Gk\delta^2 > 0$, so $1 - y \geq (G - X - Gk\delta^2)/G$, that is, $y \leq y_+ < 1$. Since f is a decreasing bijection $(0, \infty) \rightarrow (0, 1)$, the solution

$$T = \frac{\sigma(1 - y) + \sqrt{\sigma^2(1 - y)^2 + ky(1 - y)}}{ky}$$

exists, is unique, and lies in the compact interval $[f^{-1}(y_+), f^{-1}(y_-)]$.

With these inverse rates, the quantities x_j of (3) are $x_j = r_j f(t_j)$. Hence $\sum_j x_j = Sf(\delta) + (G - S)y = X$. For the lag shifts, each fast piece has $\alpha_j = r_j \delta / d_j \leq r_j \delta$, so these sum to at most $G\delta$. Each slow piece has $\alpha_j = x_j T / (1 + 2\sigma T) < x_j / (2\sigma)$, and the slow x_j sum to $X - Sf(\delta) < X(1 - f(\delta)) \leq Gk\delta^2$. By (7), $\sum_j \alpha_j < q/2$.

Replace the port inverse rate q by $q' = q - \sum_j \alpha_j \in (q/2, q)$ and keep the other core inverse rates. By (4), $L_r(z) = X - z \sum_j \alpha_j$, so

$$(zQ' - B - L_r(z)ee^T)v = (zQ - B_X)v = 0$$

for the static eigenvector v . By the remarks after (5), z is an eigenvalue of the scaled A_r . All rates are fixed core rates, a port rate in $[1/q, 2/q]$, $1/\delta$, or $1/T$ with T in a fixed compact interval. $\square$

The constants in (7) are sufficient, not necessary. The construction succeeds whenever $y \in (0, 1)$ and $\sum_j \alpha_j < q$, and both conditions can be checked directly for a given partition.

Theorem 4.3 (Exact distribution-free granularity threshold). *Let B be D -stable and let τ be defined by (2). Then $0 \leq \tau \leq -B_{pp}$, and for every finite positive partition r of G :*

- (i) *if $G < \tau$, then A_r is D -stable;*
- (ii) *if $G = \tau$, then A_r is D -stable if and only if $\det B_\tau \neq 0$. Every nonzero eigenvalue of every positive scaling has negative real part, and if $\det B_\tau = 0$ then 0 is an eigenvalue of every positive scaling;*
- (iii) *if $G > \tau$, there is $\varepsilon > 0$, independent of the number of pieces, such that every partition of G with all pieces smaller than ε has a strictly unstable positive scaling.*

Proof. The bounds are Lemma 2.1. If $G \leq \tau$, then B_X is D -stable for $0 < X < G$ by Lemma 4.1. For $G < \tau$, also $\det B_G \neq 0$ by the same lemma. Theorem 3.2 gives (i) and (ii). If $G > \tau$, the infimum (2) provides $X \in [\tau, G)$ with strict growth, and $X > 0$ because B is D -stable. Lemma 4.2 gives (iii). $\square$

Remark 4.4 (What the theorem does and does not say). Theorem 4.3 is an exact *reduction*. The dynamic question about all partitions and all rates becomes a static D -stability question along one ray. Evaluating τ is as hard as deciding D -stability of the matrices B_X . It is explicit whenever that static problem can be solved: in dimensions two and three (Section 7), in classified families (Section 8), or from certificates. The theorem also says nothing about *coarse* partitions of $G > \tau$; those are the subject of Sections 5–8.

4.1 Bounded ratios of time scales

In applications the time scales cannot be arbitrarily disparate. For a positive diagonal D let $\kappa(D) = \max_i D_{ii} / \min_i D_{ii}$ be its *spread*.

Proposition 4.5 (Bounded spread raises the threshold). *Let B be D -stable.*

- (a) *For every $G > \tau$ there are $\varepsilon > 0$ and $\kappa_G < \infty$ such that every partition of G into pieces smaller than ε has a strictly unstable scaling of spread at most κ_G .*
- (b) *Suppose $\det B_\tau \neq 0$. For every $\kappa \geq 1$ there is $\eta > 0$ such that for every $G < \tau + \eta$, every finite positive partition r of G and every positive D with $\kappa(D) \leq \kappa$, the matrix DA_r is Hurwitz.*
- (c) *If $\det(-B_G) < 0$, then every positive scaling of every partition of G has a positive real eigenvalue.*

Consequently, the time-scale-limited threshold $\tau_\kappa = \sup\{G' : \text{every partition of every } G < G' \text{ is Hurwitz at all scalings of spread } \leq \kappa\}$ is nonincreasing in κ and tends to τ as $\kappa \rightarrow \infty$. It satisfies $\tau_\kappa > \tau$ for every finite κ when $\det B_\tau \neq 0$.

Proof. (a) is Lemma 4.2(b) applied to one static witness $X \in (\tau, G)$.

(b) Suppose the claim fails for some κ . Then there are $G_k \downarrow \tau$, partitions $r^{(k)}$ and scalings of spread at most κ with eigenvalues λ_k , $\operatorname{Re} \lambda_k \geq 0$. By Theorem 4.3, $G_k > \tau$. Multiplying a scaling by a constant multiplies the spectrum by that constant, so we may normalize all inverse rates to lie in $[1/\kappa, 1]$. For $\operatorname{Re} \lambda \geq 0$ and $t > 0$ we have $|1 + \lambda t| \geq 1$. Hence $|L_{r^{(k)}}(\lambda_k)| \leq G_k$, and (5) with $\|v_k\| = 1$ gives $|\lambda_k| \leq \kappa(\|B\| + G_k)$. Represent the attachments by the measures $\mu_k = \sum_j r_j^{(k)} \delta_{t_j}$ on $[1/\kappa, 1]$, of mass G_k , so that $L_{r^{(k)}}(\lambda) = \int d\mu_k(t)/(1 + \lambda t)$. Pass to a subsequence along which $\mu_k \rightarrow \mu$ weakly, with mass τ . Along it the core inverse rates converge to $Q_* \in [1/\kappa, 1]^n$, $\lambda_k \rightarrow \lambda_*$ with $\operatorname{Re} \lambda_* \geq 0$, and $v_k \rightarrow v_*$ with $\|v_*\| = 1$. The integrands are uniformly Lipschitz in λ on $\{\operatorname{Re} \lambda \geq 0\}$, so $L_{r^{(k)}}(\lambda_k) \rightarrow L(\lambda_*) = \int d\mu/(1 + \lambda_* t)$. In the limit,

$$(\lambda_* Q_* - B - L(\lambda_*) e e^T) v_* = 0.$$

If $\lambda_* = 0$, then $L(0) = \tau$ and $B_\tau v_* = 0$, contradicting $\det B_\tau \neq 0$. If $\lambda_* \neq 0$, integrate Lemma 3.1 against μ : $L(\lambda_*) = x - \lambda_* \alpha$ with $\alpha \geq 0$ and $0 \leq x \leq \tau$. If $\mu \neq 0$ then $x < \tau$, because the integrand of $\tau - x$ is positive on $[1/\kappa, 1]$; if $\mu = 0$ then $x = 0$. In both cases B_x is D-stable, by Lemma 4.1 or because $B_0 = B$. Yet λ_* is a closed-right-half-plane eigenvalue of the positive scaling $(Q_* + \alpha e e^T)^{-1} B_x$. This is a contradiction.

(c) The characteristic polynomial of DA_r is monic, and its constant term is $\det(-DA_r) = \det D \cdot \det(-B_G) < 0$. Hence it has a positive real root.

The consequences follow from (a), (b) and monotonicity in κ . $\square$

Part (b) says that near a threshold with $\det B_\tau \neq 0$, instability is bought only with ever larger ratios of time scales. Part (c) says that a real-root crossing, where $\det(-B_X)$ changes sign at τ , is not softened at all. For the four-state example of Section 8, an explicit witness destabilizes the split $(1, 1)$ with spread $4375/4$.

5 Fixed partitions: planar geometry

Theorem 4.3 classifies all sufficiently fine partitions. We now decide a *given* partition. Order the coordinates so that the port comes first and write

$$B = \begin{pmatrix} b & a^T \\ c & C \end{pmatrix}, \quad Z(Q_c) = a^T (iQ_c - C)^{-1} c \quad (\det(iQ_c - C) \neq 0),$$

where Q_c ranges over positive diagonal inverse rates of the $n - 1$ inner coordinates. We assume $n \geq 2$. The *core envelope* is

$$\Theta(X) = \sup\{\operatorname{Im} Z(Q_c) : Q_c > 0, \det(iQ_c - C) \neq 0, \operatorname{Re} Z(Q_c) = -b - X\}, \quad \sup \emptyset = -\infty. \quad (9)$$

Thus $\Theta(X) > 0$ means that the static matrix B_X has a contact at some scaling.

5.1 The image of a partition

At frequency one, the attachments contribute $L_r(i) = \sum_j r_j/(1 + it_j) = X - iY$. For $0 < X < G$ define the *protection envelope*

$$\Lambda_r(X) = \min_{(k, S)} \sqrt{(X - r(S))(r(S) + r_k - X)}, \quad (10)$$

the minimum being over all $1 \leq k \leq m$ and $S \subseteq [m] \setminus \{k\}$ with $r(S) \leq X \leq r(S) + r_k$, where $r(S) = \sum_{j \in S} r_j$. The graph of Λ_r is the lower envelope of the semicircles with diameters $[r(S), r(S) + r_k]$. It vanishes exactly at the subset sums of r . For one piece, $\Lambda_{(G)}(X) = \sqrt{X(G - X)}$.

Lemma 5.1 (Exact attachment image). *As the inverse rates t_j range over $(0, \infty)$, the point (X, Y) with $X - iY = \sum_j r_j/(1 + it_j)$ ranges over*

$$\begin{aligned} \{(X, \sqrt{X(G - X)}) : 0 < X < G\} & \quad \text{if } m = 1, \\ \{(X, Y) : 0 < X < G, \Lambda_r(X) < Y \leq \sqrt{X(G - X)}\} & \quad \text{if } m \geq 2. \end{aligned}$$

Proof. One piece gives $x = r/(1 + t^2)$ and $y = rt/(1 + t^2) = \sqrt{x(r - x)}$. As t ranges over $(0, \infty)$, x ranges over $(0, r)$ bijectively, with inverse $t = \sqrt{(r - x)/x}$. Hence the attainable points are the images of $\varphi(x) = \sum_j \sqrt{x_j(r_j - x_j)}$ on the open slices $\Sigma_X^\circ = \{x \in \prod_j (0, r_j) : \sum_j x_j = X\}$, $0 < X < G$. By Cauchy-Schwarz, $\varphi(x) \leq \sqrt{\sum_j x_j} \sqrt{\sum_j (r_j - x_j)} = \sqrt{X(G - X)}$, with equality at $x_j = r_j X/G$, which lies in Σ_X° . The function φ is strictly concave on the closed slice Σ_X , which is a polytope. Hence its minimizers over Σ_X are vertices. At a vertex at most one coordinate x_k is strictly between its bounds; the others equal r_j ($j \in S$) or 0. The vertex values are the numbers in (10), so $\min_{\Sigma_X} \varphi = \Lambda_r(X)$. If $m \geq 2$, every vertex has a coordinate at a bound, so vertices are not in Σ_X° and $\varphi > \Lambda_r(X)$ there. The image of the convex set Σ_X° under the continuous φ is an interval. Its infimum is $\Lambda_r(X)$, because Σ_X° is dense in Σ_X , and its maximum $\sqrt{X(G - X)}$ is attained. $\square$

5.2 The fixed-partition test

Theorem 5.2 (Exact fixed-partition test). *Let B be D -stable with $n \geq 2$. Then A_r is D -stable if and only if*

$$\det(-B_G) > 0 \quad \text{and} \quad \Theta(X) \leq \Lambda_r(X) \quad \text{for all } 0 < X < G. \quad (11)$$

Proof. Zero eigenvalues. $\det(-DA_r) = \det D \det(-B_G)$. A Hurwitz matrix M has $\det(-M) > 0$, so the determinant condition is necessary. It excludes zero eigenvalues.

Contacts. After frequency normalization, a contact is a singular pencil $i\hat{Q} - A_r$. Since $1 + it_j \neq 0$, this means $\det(iQ - B - L_r(i)ee^T) = 0$ with $Q = \text{diag}(q_h, Q_c)$. Suppose first $\det(iQ_c - C) \neq 0$. The Schur complement on the port gives

$$\det(iQ - B - Lee^T) = \det(iQ_c - C) (iq_h - b - L - Z(Q_c)).$$

With $L = X - iY$, singularity means $X = -b - \text{Re } Z$ and $q_h = \text{Im } Z - Y$. Given Q_c , some positive q_h and some attachment rates realize this if and only if $X \in (0, G)$ and some attainable height Y at X satisfies $Y < \text{Im } Z$. By Lemma 5.1, this happens if and only if $\text{Im } Z > \Lambda_r(X)$: for $m \geq 2$ the attainable heights are exactly $(\Lambda_r(X), \sqrt{X(G - X)})$, and for $m = 1$ the only height is $\Lambda_{(G)}(X)$. Taking the supremum over Q_c , contacts with a regular inner pencil exist if and only if $\Theta(X) > \Lambda_r(X)$ for some $X \in (0, G)$.

Singular inner pencils give no contacts. If $\det(iQ_c - C) = 0$, expansion along the port gives $\det(iQ - B - Lee^T) = -a^T \text{adj}(iQ_c - C)c$, independent of L and q_h . With $L = 0$ this is $\det(iQ - B) \neq 0$, because B is D -stable.

A stable base point. Fix any positive core rates D_c and give every attachment rate ϵ . At $\epsilon = 0$ the scaled matrix is block triangular with eigenvalues those of $D_c B$, which lie in the open left half-plane, and 0 with multiplicity m . For small $\epsilon > 0$, m eigenvalues lie in a small disk around 0. The value $-\epsilon$ is an eigenvalue of multiplicity at least $m - 1$, with eigenvectors

$x = 0$, $\sum_j r_j w_j = 0$. The remaining eigenvalue λ_s in the disk is real, since its conjugate is also in the disk. Comparing determinants,

$$\lambda_s(-\epsilon)^{m-1} \prod (\text{fast eigenvalues}) = \det(DA_r) = \epsilon^m \det D_c (-1)^m \det B_G,$$

and the product of the fast eigenvalues tends to $\det(D_c B)$. Hence

$$\lambda_s = -\epsilon \frac{\det(-B_G)}{\det(-B)} (1 + o(1)) < 0,$$

and the scaled matrix is Hurwitz for small ϵ .

Conclusion. If (11) holds, no eigenvalue of any positive scaling lies on the imaginary axis. The positive diagonal cone is connected and eigenvalues depend continuously on the rates. Starting from the Hurwitz base point, no eigenvalue can reach the closed right half-plane. Conversely, D-stability excludes contacts and forces the determinant sign. $\square$

The base-point computation also shows the role of the determinant condition: when $\det(-B_G) < 0$, the slow eigenvalue is positive.

Corollary 5.3 (Disk form). *For $1 \leq k \leq m$ and $S \subseteq [m] \setminus \{k\}$ let $\mathbb{D}_{k,S}$ be the closed disk in $\mathbb{C}$ with diameter $[-b - r(S) - r_k, -b - r(S)]$ on the real axis. Then A_r is D-stable if and only if $\det(-B_G) > 0$ and every core value $Z = Z(Q_c)$ with $\text{Im } Z > 0$ and $-b - G < \text{Re } Z < -b$ lies in every disk $\mathbb{D}_{k,S}$ whose diameter contains $\text{Re } Z$. For a single piece this is one disk, with diameter $[-b - G, -b]$.*

Proof. With $X = -b - \text{Re } Z$ and $s = r(S)$, the inequality $(\text{Im } Z)^2 \leq (X - s)(s + r_k - X)$ is $(\text{Re } Z + b + s + r_k/2)^2 + (\text{Im } Z)^2 \leq (r_k/2)^2$. The condition $\text{Im } Z \leq \Lambda_r(X)$ requires this for every (k, S) with $s \leq X \leq s + r_k$. $\square$

In the Z -plane, a lump offers one large disk. A fragmented load replaces it by a chain of small disks that touch the real axis at every subset sum. Near a subset sum no core value with positive imaginary part is tolerated. Figure 3 shows the same picture in the coordinates $(X, \text{Im } Z)$, where disks become semicircles.

Remark 5.4 (Negative loads). For negative loads $r_j < 0$ with total $G < 0$, both X and Y in $L_r(i) = X - iY$ are negative. A contact requires $q_h = \text{Im } Z - Y = \text{Im } Z + |Y| > 0$. Now the largest attainable $|Y|$ at fixed X is what matters. By the Cauchy–Schwarz step of Lemma 5.1 it equals $\sqrt{|X|(|G| - |X|)}$ and is attained with synchronized times. It depends on the total only. The proof of Theorem 5.2 therefore shows that, for damping partners, D-stability is independent of the partition. Granularity is a positive-feedback phenomenon: for positive loads the protective height is the *minimum* over the partition’s semicircles.

6 Subset sums, edges and corners

The planar test has three practical consequences. The first reduces every fixed-partition verdict to one-attachment systems and static matrices.

Lemma 6.1 (Edges and corners). *Let B be D-stable and let r be a partition of G . Suppose that*

- (i) $\det(-B_G) > 0$;
- (ii) *for every subset S with $0 < r(S) < G$, the static corner $B_{r(S)}$ is D-stable;*
- (iii) *for every k and $S \subseteq [m] \setminus \{k\}$, the one-attachment edge system, with core $B_{r(S)}$ and a single piece of load r_k , is D-stable.*

Then A_r is D -stable. Conversely, if A_r is D -stable then (i)–(iii) hold. There are at most $m2^{m-1}$ edges and 2^m corners; equal pieces reduce these counts.

Proof. Suppose $\Theta(X) > \Lambda_r(X)$ for some $X \in (0, G)$. Then some core value Z has $\operatorname{Re} Z = -b - X$ and $\operatorname{Im} Z > \Lambda_r(X)$. Choose (k, S) attaining the minimum in (10), and put $s = r(S)$. The core B_s has the same inner block and port self-coefficient $b + s$, so it has the same core values Z . If $s < X < s + r_k$, then $X' = X - s \in (0, r_k)$ satisfies $\operatorname{Re} Z = -(b + s) - X'$ and $\operatorname{Im} Z > \sqrt{X'(r_k - X')}$. By Theorem 5.2 applied to the edge, the edge has a contact, contradicting (iii). If X is an endpoint, then $X = s'$ is a subset sum in $(0, G)$ and $\Lambda_r(X) = 0 < \operatorname{Im} Z$. The static matrix $B_{s'}$ then has a contact with port inverse rate $q_h = \operatorname{Im} Z$, contradicting (ii). Hence (11) holds.

For the converse, let A_r be D -stable. Then (i) holds by Theorem 5.2. The function $X \mapsto \det(-B_X)$ is affine and positive at $X = 0$ and $X = G$, hence positive on $[0, G]$. Let s' be a subset sum in $(0, G)$. By Proposition 6.2, $B_{s'}$ has no strict growth. Since $\Theta(s') \leq \Lambda_r(s') = 0$, no core value with $\operatorname{Re} Z = -b - s'$ has positive imaginary part, so $B_{s'}$ has no contact through a regular inner pencil. Singular inner pencils give none either (proof of Theorem 5.2). Together with $\det(-B_{s'}) > 0$, this makes the corner $B_{s'}$ D -stable, proving (ii). For an edge (k, S) with $s = r(S)$, the core B_s is D -stable (it is B or a corner), its determinant condition is $\det(-B_{s+r_k}) > 0$, and its core envelope at $X' \in (0, r_k)$ is $\Theta(s + X') \leq \Lambda_r(s + X') \leq \sqrt{X'(r_k - X')}$. Theorem 5.2 applied to the edge gives (iii). $\square$

The second consequence identifies the mechanism: fast pieces act as static load and slow pieces vanish.

Proposition 6.2 (Subset-sum obstruction). *Let B be any real matrix and r a partition. If for some $S \subseteq [m]$ the static matrix $B_{r(S)}$ has a strictly unstable positive scaling, then A_r has a strictly unstable positive scaling. This holds in particular for $S = \emptyset$ (B itself) and for $S = [m]$ (B_G).*

Proof. Let $B_{r(S)}$ have eigenvalue z_0 with $\operatorname{Re} z_0 = \sigma_0 > 0$ at inverse rates Q . Give the pieces in S inverse rate ϵ and the others inverse rate $1/\epsilon$. On the disk $K = \{|z - z_0| \leq \rho\}$ with $\rho < \sigma_0/2$ we have $\operatorname{Re} z \geq \sigma_0/2$. Using $|1 + zt| \geq 1$ and $|1 + z/\epsilon| \geq 1 + \sigma_0/(2\epsilon)$,

$$\left| L_\epsilon(z) - r(S) \right| \leq \sum_{j \in S} \frac{r_j |z| \epsilon}{|1 + z\epsilon|} + \sum_{j \notin S} \frac{r_j}{|1 + z/\epsilon|} \leq G \left(|z| \epsilon + \frac{2\epsilon}{\sigma_0} \right),$$

uniformly on K . Hence $F_\epsilon(z) = \det(zQ - B - L_\epsilon(z)ee^T)$ converges uniformly on K to $F_0(z) = \det(zQ - B_{r(S)})$. Choose ρ so that F_0 has no zero on ∂K ; F_0 is a nonzero polynomial. By Hurwitz's theorem [8], F_ϵ has a zero in K for small ϵ . By (5) it is an eigenvalue of the scaled A_r with positive real part. $\square$

The third consequence is monotonicity. A partition r' *refines* r if the pieces of r' can be grouped so that the group sums are the pieces of r .

Corollary 6.3 (Merging is safe). *If r' refines r , then every eigenvalue of a positive scaling of A_r is an eigenvalue of a positive scaling of $A_{r'}$. In particular, D -stability of $A_{r'}$ implies D -stability of A_r , and a strictly unstable scaling of A_r yields one for $A_{r'}$. The single lump (G) is the most protective partition: if any partition of G is D -stable, so is $A_{(G)}$.*

Proof. Give the pieces of r' in each group the rate of the corresponding piece of r . Then $L_{r'} = L_r$, and the characteristic polynomials satisfy $\chi_{A_{r'}} = \chi_{A_r} \prod (s + d_\ell)^{\nu_\ell - 1}$. Here group ℓ has ν_ℓ pieces and rate d_ℓ , and the extra factors are the difference modes inside the groups, all with eigenvalue $-d_\ell < 0$. For the last statement, note that (G) is refined by every partition. $\square$

Taken together, Theorem 4.3, Proposition 6.2 and Corollary 6.3 describe the landscape at a fixed total. The lump is safest; merging pieces helps. A partition is doomed as soon as a subset sum enters the static instability window. In the limit of fine dust, subset sums are dense and only τ matters. Whether the subset-sum criterion is also *sufficient* depends on the core. By Lemma 6.1 it is sufficient whenever, for the partitions in question, all corners are D-stable as soon as they have no strict growth, and every edge is D-stable as soon as its two static corners are. Section 8 gives a family where this holds.

7 Small cores and the minimum dimension

A granularity effect at fixed total requires the static ray to lose and then regain stability. We show that this cannot happen in dimension at most three.

Two states. Let $B = \begin{pmatrix} -a & p \\ q & -b \end{pmatrix}$ be D-stable ($a, b \geq 0$, $a + b > 0$, $t = ab - pq > 0$) and load coordinate one. A real 2×2 matrix has no strict growth exactly when its diagonal entries are nonpositive and its determinant is nonnegative. Along the ray these conditions read $a - X \geq 0$ and $t - bX \geq 0$, so

$$\tau = \begin{cases} \min\{a, t/b\}, & b > 0, \\ a, & b = 0, \end{cases}$$

and threshold equality is safe exactly when $\det(-B_\tau) = t - b\tau > 0$. Both conditions fail permanently once they fail, so there is no reentry.

Three states. For a 3×3 matrix M let $a_i = -M_{ii}$, let u_{ij} be the principal 2×2 minors of $-M$, and $t = \det(-M)$. The characteristic polynomial of DM is $\det(sI - DM) = s^3 + c_1s^2 + c_2s + c_3$ with $c_1 = \sum a_i d_i$, $c_2 = \sum u_{ij} d_i d_j$ and $c_3 = t d_1 d_2 d_3$. Its roots lie in the closed left half-plane exactly when all $c_i \geq 0$ and $c_1 c_2 \geq c_3$ [15]. Dividing by $d_1 d_2 d_3$, the infimum of $c_1 c_2 / (d_1 d_2 d_3)$ over $d > 0$ is $(\sqrt{P_1} + \sqrt{P_2} + \sqrt{P_3})^2$, where $P_1 = a_1 u_{23}$, $P_2 = a_2 u_{13}$ and $P_3 = a_3 u_{12}$. Indeed, the difference is a sum of three squares of the form $(\sqrt{a_1 u_{13} d_1 / d_2} - \sqrt{a_2 u_{23} d_2 / d_1})^2$. Hence M has no strict growth if and only if all $a_i, u_{ij}, t \geq 0$ and $t \leq (\sqrt{P_1} + \sqrt{P_2} + \sqrt{P_3})^2$; compare Cain's characterization of 3×3 D-stability [6]. For positive data the infimum is attained, because the three conditions making the squares vanish are consistent. The quantity $c_1 c_2 / (d_1 d_2 d_3)$ is monotone in the data, so data with zeros are covered by raising every datum by $\eta > 0$ and letting $\eta \rightarrow 0$. Now load coordinate one of a D-stable B . Write a, b, c for the diagonal of $-B$, u, v, w for its principal minors on $\{1, 2\}, \{1, 3\}, \{2, 3\}$, and $t = \det(-B)$. Along the ray, the data are $a - X, b, c, u - bX, v - cX, w$ and $t - wX$, so $P_1 = A := w(a - X)$, $P_2 = E_2 := b(v - cX)$, $P_3 = E_3 := c(u - bX)$ and $T := t - wX$ are affine in X . Let

$$L = \min(\{a\} \cup \{u/b : b > 0\} \cup \{v/c : c > 0\} \cup \{t/w : w > 0\}).$$

Beyond L some principal minor of $-B_X$ is negative, and this forces strict growth: scale the complementary coordinates towards zero.

Proposition 7.1 (No reentry in dimension at most three). *If $n \leq 3$, the set of $X \geq 0$ at which B_X has no strict growth is the interval $[0, \tau]$. Consequently, for $G > \tau$ every partition of G has a strictly unstable positive scaling. For every G , the D-stability of A_r depends only on G . Granularity sensitivity at a fixed total therefore requires at least four core states. Section 8 shows that four suffice.*

Proof. For $n = 1$ this is trivial, and $n = 2$ is shown above. For $n = 3$, on $[0, L]$ the gap $g(X) = (\sqrt{A} + \sqrt{E_2} + \sqrt{E_3})^2 - T$ equals the affine function $A + E_2 + E_3 - T$ plus $2(\sqrt{AE_2} + \sqrt{AE_3} + \sqrt{E_2 E_3})$. The geometric mean of two nonnegative affine functions is

concave, so g is concave on $[0, L]$ with $g(0) \geq 0$. Its nonnegative set is therefore an interval $[0, X_1]$, and there is strict growth on $(X_1, L]$ and beyond L . Thus the growth-free set is $[0, \tau]$ with $\tau = \min(X_1, L)$. If $G > \tau$, then B_G has strict growth, and Proposition 6.2 with $S = [m]$ applies. If $G \leq \tau$, Theorem 4.3 decides every partition in the same way. $\square$

The threshold is also algebraically explicit in three states. Let $E = T - A - E_2 - E_3$ and $V = AE_2 + AE_3 + E_2E_3$. The polynomial

$$\mathcal{P} = E^4 - 8VE^2 - 64AE_2E_3E + 16\{V^2 - 4AE_2E_3(A + E_2 + E_3)\}$$

is the product of $T - (\sqrt{A} \pm \sqrt{E_2} \pm \sqrt{E_3})^2$ over the four sign choices. For the ray data it has degree at most two in X : the slopes satisfy $A' = T' = -w$ and $E'_2 = E'_3 = -bc$, and the quartic and cubic terms cancel. For general affine data the degree can be four. If $\mathcal{P} \equiv 0$ then $\tau = L$. Otherwise τ is either L or a root of $\mathcal{P}$ in $[0, L)$, and $\tau = 0$ can occur even for D-stable B ; testing the radical inequality between consecutive roots selects it, and concavity guarantees a single transition.

8 An exactly solvable four-state family

Consider the rational family

$$B(h) = \begin{pmatrix} -1 & -7/10 & -7/10 & -14/5 \\ 4 & -1 & -7/10 & -14/5 \\ 4 & 4 & -1 & -14/5 \\ -1 & -1 & -1 & -h \end{pmatrix}, \quad h \geq 4, \quad (12)$$

with the port at coordinate four. Loading the port moves along the family: $B(h) + Xe_4e_4^T = B(h - X)$. Put

$$K(h) = 3890h^2 - 38717h + 93492, \quad h_{\pm} = \frac{38717 \pm 329\sqrt{409}}{7780},$$

so that $h_- \approx 4.121258$ and $h_+ \approx 5.831698$.

Proposition 8.1 (Static classification). *For $h \geq 4$, $B(h)$ is D-stable if and only if $K(h) \geq 0$, that is, if and only if $h \leq h_-$ or $h \geq h_+$. For $h_- < h < h_+$ it has a strictly unstable positive scaling. Moreover $\det(-B(h)) = \frac{466}{25}h - \frac{18613}{250} > 0$ for $h \geq 4$.*

Proof. Let the core rates be x, y, z and the port rate v . Put $p = x + y + z$, $q = xy + xz + yz$, $r = xyz$, $U = pq - 9r = x(y - z)^2 + y(x - z)^2 + z(x - y)^2 \geq 0$ and $V = q^2 - 3pr \geq 0$. The characteristic polynomial $\det(sI - \text{diag}(x, y, z, v)B(h)) = s^4 + c_1s^3 + c_2s^2 + c_3s + c_4$ has coefficients

$$\begin{aligned} c_1 &= p + hv, & c_2 &= \frac{19}{5}q + (h - \frac{14}{5})vp, \\ c_3 &= \frac{466}{25}r + (\frac{19}{5}h - \frac{371}{25})vq, & c_4 &= (\frac{466}{25}h - \frac{18613}{250})vr, \end{aligned}$$

all positive for $h \geq 4$. The third Hurwitz determinant is $c_1c_2c_3 - c_3^2 - c_1^2c_4 = \sum_{j=0}^3 F_jv^j$ with

$$\begin{aligned} F_0 &= \frac{466}{625}r(95U + 389r), \\ 1250F_1 &= (18050h - 70490)qU + (73910h + 57134)qr + 27825p^2r, \\ 625F_2 &= 5(5h - 14)(95h - 371)pU + 371(95h - 371)V + (9725h^2 + 22855h - 179193)pr, \\ 250F_3 &= h\{2(5h - 14)(95h - 371)U + K(h)r\}. \end{aligned}$$

For $h \geq 4$ every factor other than $K(h)$ is positive, and $F_0 > 0$. If $K(h) \geq 0$ the determinant Δ_3 is positive at every rate. Since $\Delta_3 = c_3\Delta_2 - c_1^2c_4$ with $c_i > 0$, also $\Delta_2 > 0$, and $\Delta_4 = c_4\Delta_3 > 0$. Routh–Hurwitz gives stability. If $K(h) < 0$, take $x = y = z = 1$, so that $U = 0$ and the leading coefficient in v is $hK(h)/250 < 0$. For large v the third Hurwitz determinant is negative, so the scaled matrix is not Hurwitz. By Orlando’s formula [15], the third Hurwitz determinant of a monic quartic is, up to sign, the product of the pairwise sums of its roots. It therefore vanishes whenever there is an imaginary-axis pair, and $c_4 > 0$ excludes a zero root. Hence some root has positive real part. $\square$

For $0 \leq X \leq h_0 - 4$, the static window of $B(h_0)$ is therefore $X \in (h_0 - h_+, h_0 - h_-)$. For $h_0 = 6$ it is $(\tau, 6 - h_-) \approx (0.1683, 1.8787)$ with $\tau = 6 - h_+ = (7963 - 329\sqrt{409})/7780$. The matrix $B(6)$ is D-stable, and so is its fully loaded version $B(4)$, but the middle of the ray is strictly unstable.

Lemma 8.2 (One-attachment edges). *Let $4 \leq J < H$ with $K(H) \geq 0$ and $K(J) \geq 0$. The system with core $B(H)$ and a single attachment of load $g = H - J$ at the port is D-stable.*

Proof (exact computer-assisted certificate). With attachment rate w the characteristic polynomial of the scaled matrix is $\det(sI - DA) = s^5 + \sum_{i=1}^5 a_i s^{5-i}$ with

$$\begin{aligned} a_1 &= c_1 + w, & a_2 &= c_2 + w[p + Jv], & a_3 &= c_3 + w\left[\frac{19}{5}q + \left(J - \frac{14}{5}\right)vp\right], \\ a_4 &= c_4 + w\left[\frac{466}{25}r + \left(\frac{19}{5}J - \frac{371}{25}\right)vq\right], & a_5 &= w\left(\frac{466}{25}J - \frac{18613}{250}\right)vr, \end{aligned}$$

where c_i are the coefficients of Proposition 8.1 at $h = H$. Only the fast-end damping $J = H - g$ enters the w -terms. The coefficients are symmetric in x, y, z , so we may assume $x \geq y \geq z$ and substitute $x = \alpha + \beta + \gamma$, $y = \beta + \gamma$, $z = \gamma$ with $\alpha, \beta \geq 0$ and $\gamma, v, w > 0$. The Hurwitz determinants $\Delta_1, \dots, \Delta_4$ then become polynomials in $(\alpha, \beta, \gamma, v, w)$ with 702 monomials in total, whose coefficients are polynomials in (H, J) . Every coefficient is nonnegative on the region $\mathcal{R} = \{4 \leq J \leq H, K(H) \geq 0, K(J) \geq 0\}$:

- 664 become polynomials with nonnegative coefficients after $J = 4 + j$, $H = 4 + j + g$ ($j, g \geq 0$);
- 12 are $K(H)$ or $K(J)$ times such a polynomial;
- 8 depend on H only, are positive at $H = 4$ with positive leading coefficient, and have all real roots ≥ 4 in $(41213/10000, 58316/10000) \subset (h_-, h_+)$;
- 18 are nonnegative on each of the three pieces of $\mathcal{R}$: on $\{4 \leq J \leq H \leq h_-\}$ via barycentric coordinates on the exact triangle with vertices $(4, 4), (4, h_-), (h_-, h_-)$; on $\{h_+ \leq J \leq H\}$ via $J = h_+ + j$, $H = h_+ + j + g$, with coefficients $a + b\sqrt{409}$ ($a, b \in \mathbb{Q}$) whose signs are decided exactly by comparing a^2 with $409b^2$; and on $\{4 \leq J \leq h_-, H \geq h_+\}$ via $H = 58316/10000 + g$, $J = 4 + \frac{1213}{10000}s/(1+s)$ after clearing denominators. In each case all coefficients are nonnegative.

Each Δ_k has a monomial free of α, β whose coefficient is positive on all of $\{4 \leq J \leq H\}$, so $\Delta_k > 0$. Finally $\Delta_5 = a_5\Delta_4 > 0$, because $\frac{466}{25}J > \frac{18613}{250}$ for $J \geq 4$. By the Hurwitz criterion the scaled system is Hurwitz for all positive rates. The computation is exact rational and algebraic arithmetic, reproduced by the script `check_paper.py` (Section 11). $\square$

Theorem 8.3 (Exact classification of all partitions). *Let $h_0 \geq 4$ and $0 < G \leq h_0 - 4$, and attach a finite positive partition r of G at the port of $B(h_0)$. Then A_r is D-stable if and only if no subset sum $r(S)$, $S \subseteq [m]$, lies in the open window $(h_0 - h_+, h_0 - h_-)$. Equivalently, $K(h_0 - r(S)) \geq 0$ for all S . Otherwise A_r has a strictly unstable positive scaling.*

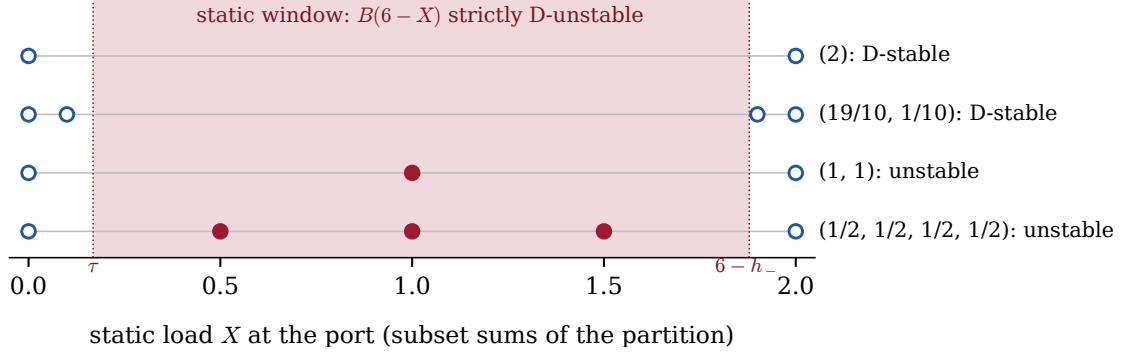


Figure 2: Theorem 8.3 for the core $B(6)$ at total load 2. The shaded interval is the static window where $B(6 - X)$ is strictly D-unstable. Dots are the subset sums of each partition; a filled dot inside the window certifies instability (Proposition 6.2). Partitions without such a dot are D-stable (Theorem 8.3).

Proof. If some subset sum lies in the window, then $B(h_0 - r(S))$ is strictly unstable by Proposition 8.1, and Proposition 6.2 applies. The case $S = \emptyset$ covers a core that is not itself D-stable. Conversely, suppose that all $K(h_0 - r(S)) \geq 0$. Every corner $B(h_0 - r(S))$ is D-stable by Proposition 8.1, since $h_0 - r(S) \geq h_0 - G \geq 4$. Every edge has core $B(H)$ with $H = h_0 - r(S)$ and a single piece r_k , hence fast-end damping $J = h_0 - r(S \cup \{k\}) \geq 4$. Both $K(H)$ and $K(J)$ are nonnegative, so the edge is D-stable by Lemma 8.2. Finally $\det(-B(h_0 - G)) > 0$. By Lemma 6.1, A_r is D-stable. $\square$

Corollary 8.4 (Load two on $B(6)$). *Attach a partition r of $G \in (0, 2]$ to the port of $B(6)$.*

- (i) *If $G \leq \tau = 6 - h_+ \approx 0.1683$, then A_r is D-stable.*
- (ii) *If $\tau < G < 6 - h_- \approx 1.8787$, then every partition has a strictly unstable scaling.*
- (iii) *If $6 - h_- \leq G \leq 2$, then A_r is D-stable if and only if its largest piece satisfies*

$$\max_j r_j \geq 6 - h_- = \frac{7963 + 329\sqrt{409}}{7780} \approx 1.878742.$$

Proof. (i) All subset sums are at most $G \leq \tau$. (ii) G itself is a subset sum in the window. (iii) Let $\ell = \max_j r_j$. If $\ell \geq 6 - h_-$, the sums containing the largest piece are at least $6 - h_-$. The others are at most $G - \ell \leq 2 - (6 - h_-) = h_- - 4 \approx 0.1213 < \tau$. So no sum lies in the window. If $\tau < \ell < 6 - h_-$, the largest piece is itself in the window. If $\ell \leq \tau$, the partial sums in any order increase in steps at most τ from 0 to $G \geq 6 - h_-$. The first partial sum exceeding τ is then at most $2\tau < 6 - h_-$, so it lies in the window. $\square$

At total two the verdict thus depends on the split alone, and sharply. The single site class (2) and the split (19/10, 1/10) are D-stable for all kinetics, as is (47/25, 3/25) with its largest piece 1.88. The split (15/8, 1/8), with largest piece 1.875, is not: its sum 15/8 lies in the window. Neither are (1, 1) and its refinement $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Table 1 summarizes these verdicts.

An explicit moderate-spread witness. Proposition 6.2 gives existence only; its rates are extreme. For (1, 1) an explicit rational witness is

$$D = \text{diag}(\frac{617}{1000}, \frac{617}{1000}, \frac{617}{1000}, 35, \frac{158}{5}, \frac{4}{125})$$

Table 1: Exact verdicts at total load 2 on $B(6)$.

Partition	Subset sums in $(0, 2)$	Verdict	Reason
(2)	none	D-stable	Theorem 8.3
$(19/10, 1/10)$	$1/10, 19/10$	D-stable	Theorem 8.3
$(47/25, 3/25)$	$3/25, 47/25$	D-stable	Theorem 8.3
$(15/8, 1/8)$	$1/8, 15/8$	strictly unstable scaling	$15/8$ in window
$(1, 1)$	1	strictly unstable scaling	explicit witness below
$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$	$\frac{1}{2}, 1, \frac{3}{2}$	strictly unstable scaling	refinement of $(1, 1)$

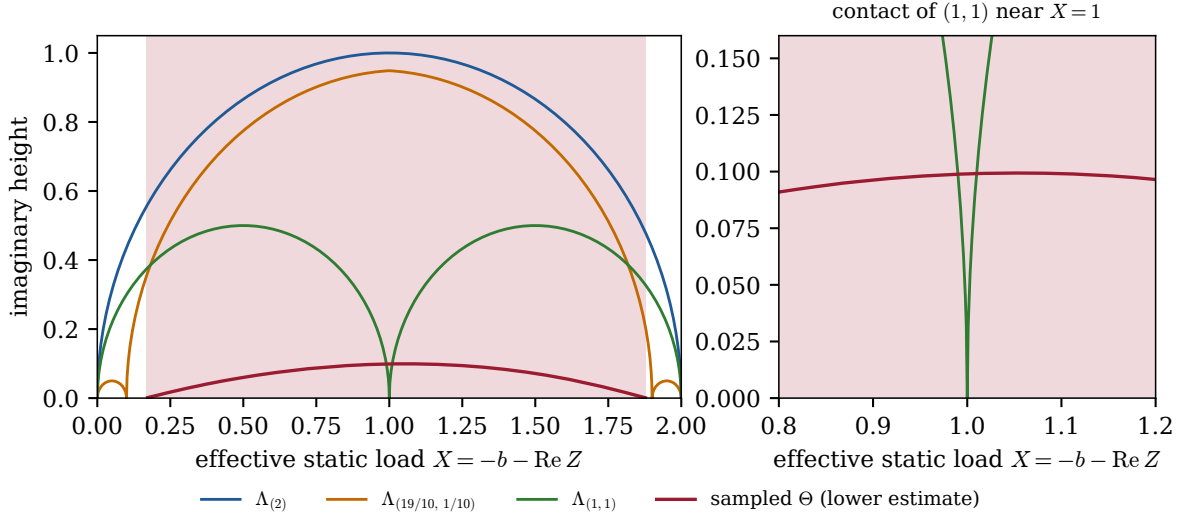


Figure 3: Planar picture for $B(6)$ at total load 2. Curves show the protection envelopes Λ_r of Lemma 5.1 for the partitions (2) , $(19/10, 1/10)$ and $(1, 1)$, and a sampled lower estimate of the core envelope Θ of (9), obtained from 6×10^5 inner rate triples. The sampled Θ is positive on the static window, shaded, and nowhere else, as it must be (Section 9). The $(1, 1)$ envelope vanishes at the subset sum $X = 1$, where the core envelope is about 0.1 (right panel), so a contact exists. The sampled curve is an illustration only; the verdicts are proved by exact certificates.

(three inner core rates, the port rate, and the two attachment rates). The characteristic polynomial of $DA_{(1,1)}$ has coefficients, in descending order,

$$1, \frac{243483}{1000}, \frac{29034069433}{5000000}, \frac{63896127507669}{12500000000}, \\ \frac{36179911928121089}{625000000000}, \frac{76473766154601901}{15625000000000}, \frac{3507069622203}{3906250000000}.$$

Its exact Routh first column has signs $+, +, +, +, -, +, +$ with no zero pivot, so there are exactly two roots with positive real part, numerically $0.0021 \pm 1.0002i$. The spread is $\kappa(D) = 4375/4 = 1093.75$. The mechanism is visible in the rates: one attachment is about a thousand times faster than the other. The fast one acts as a static load of about 1, which lies in the middle of the window, while the slow one supplies the lag. Figure 3 shows the corresponding contact in the planar picture. By Corollary 6.3, equal rates within the halves give the same two unstable roots for $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, plus two extra eigenvalues $-158/5$ and $-4/125$.

9 Several ports, reciprocal modules and hubs

9.1 Several coordinate ports

Let $\mathcal{P}$ be a set of distinct coordinates. At each $p \in \mathcal{P}$ attach a partition with total $G_p > 0$. Put $B_X = B + \sum_p X_p e_p e_p^T$ and let $\mathcal{U} = \prod_p (0, G_p)$ be the open box.

Theorem 9.1 (Open-box criterion). *Let B be D -stable. Every choice of finite positive partitions with totals (G_p) is D -stable if and only if no B_X with $X \in \mathcal{U}$ has a strictly unstable positive scaling and $\det B_G \neq 0$.*

Proof. No strict growth on $\mathcal{U}$ implies D -stability on $\mathcal{U}$. Fix positive inverse rates Q and consider $f(X) = \det(iQ - B_X)$. Each X_k enters only the (k, k) entry, so f is affine in each coordinate separately: $f = f|_{X_k=0} + X_k \partial_k f$, and $\partial_k f = -c_k$ is minus the (k, k) cofactor, which does not involve X_k . Suppose $f(X^0) = 0$ with $X^0 \in \mathcal{U}$. If some $\partial_k f(X^0) \neq 0$, the argument of Lemma 4.1 at port k produces strict growth at loads that differ from X^0 only in the k -th coordinate by an arbitrarily small amount. These loads lie in $\mathcal{U}$, a contradiction. Hence all partials vanish at every zero in $\mathcal{U}$. The restriction of f to a coordinate line through X^0 is affine with value and slope zero, so f vanishes on the whole segment of that line in $\mathcal{U}$. Each point reached is again a zero with vanishing partials. Moving one coordinate at a time reaches every point of the box, so $f \equiv 0$ on $\mathcal{U}$, and f is identically zero as a polynomial. Then $f(0) = \det(iQ - B) = 0$, contradicting D -stability of B . At zero frequency, $f_0(X) = \det(-B_X) \geq 0$ on $\mathcal{U}$: a negative value would give a positive real eigenvalue of $Q^{-1}B_X$. A nonnegative affine function of one coordinate that vanishes at an interior point has zero slope, and the same propagation gives $\det(-B) = 0$, a contradiction.

Sufficiency. Apply Lemma 3.1 independently at each port. A nonzero closed-right-half-plane eigenvalue becomes an eigenvalue of a positive scaling of some B_X , $X \in \mathcal{U}$, which is excluded. Zero is excluded by $\det(-A) = \det(-B_G) \neq 0$.

Necessity. If some B_X , $X \in \mathcal{U}$, has strict growth, apply Lemma 4.2 at every port, subtracting the respective shifts from the port inverse rates. This gives sufficiently fine partitions that realize the same eigenvalue. If $\det B_G = 0$, every choice is singular. $\square$

No stability assumption on the faces of the box is needed. Strict growth on a face persists at nearby interior points, because strict growth is an open condition.

9.2 Reciprocal relaxation modules

Corollary 9.2 (Reciprocal relaxation safety). *Let a module with state ζ obey $\dot{\zeta} = D_\zeta(M\zeta + wx_p)$ and return $\alpha w^T \zeta$ to the port, where $M = M^T$ is negative definite and $\alpha > 0$. For every positive D_ζ the module acts on the port as a partition with total $G = \alpha w^T (-M)^{-1} w$, which is independent of D_ζ . Theorem 3.2 and part (i) of Theorem 4.3 therefore hold with this total, for any number of such modules at one port. The sufficiency part of Theorem 9.1 holds for modules at several ports. The necessity part does not carry over, because it needs arbitrarily fine partitions.*

Proof. Let $S = D_\zeta^{1/2} M D_\zeta^{1/2}$, which is negative definite, and $u = D_\zeta^{1/2} w$. Eliminating the module at a candidate eigenvalue λ gives the port term $H(\lambda) = \alpha w^T (\lambda I - D_\zeta M)^{-1} D_\zeta w = \alpha u^T (\lambda I - S)^{-1} u$. By the spectral theorem,

$$H(\lambda) = \sum_\ell \frac{a_\ell}{\lambda + \lambda_\ell} = \sum_\ell \frac{g_\ell}{1 + \lambda/\lambda_\ell}, \quad a_\ell \geq 0, \lambda_\ell > 0, g_\ell = a_\ell/\lambda_\ell,$$

with $\sum_\ell g_\ell = H(0) = \alpha w^T (-M)^{-1} w$. This is exactly $L_r(\lambda)$ for the partition (g_ℓ) with inverse rates $1/\lambda_\ell$; zero residues are omitted. Its poles lie in the open left half-plane, so the elimination is

valid in the closed right half-plane, and the proofs of Theorems 3.2 and 9.1 apply verbatim. The zero-frequency identity $\det(-A) = \det(-M) \det(-B_G)$ with $\det(-M) > 0$ handles $\lambda = 0$. $\square$

More generally, if H_0 is a positive diagonal symmetrizer, meaning that $H_0 M$ is symmetric negative definite, the reciprocal return is $\alpha w^T H_0 \zeta$. The corollary is a *safety* statement. Its converse would require tuning the modal loads g_ℓ independently, which a fixed indivisible module does not allow.

9.3 A hub joining D-stable modules

The planar test has a natural generalization. Let a scalar hub with self-coefficient b be joined to modules M_j with inputs w_j (from the hub) and returns c_j^T (to the hub), and no direct coupling between modules. Assume every M_j is D-stable, and put

$$V_j = \{c_j^T (iQ_j - M_j)^{-1} w_j : Q_j > 0 \text{ diagonal}\}, \quad G_j = c_j^T (-M_j)^{-1} w_j.$$

Theorem 9.3 (Hub value-set criterion). *The hub-and-module matrix is D-stable if and only if*

$$-b - \sum_j G_j > 0 \quad \text{and} \quad \left(\sum_j V_j \right) \cap \{-b + iq : q > 0\} = \emptyset.$$

Proof. Each module pencil $iQ_j - M_j$ is invertible because M_j is D-stable. After frequency normalization, eliminating the modules turns a contact into $iq_h - b - \sum_j H_j = 0$ with $H_j \in V_j$; a zero hub component would force every module component to vanish. Since $q_h > 0$ is free, contacts are exactly the points of the Minkowski sum on the stated ray. At zero, $\det(-A) = \prod_j \det(-M_j) (-b - \sum_j G_j)$ with every $\det(-M_j) > 0$, so the first condition is necessary and excludes zero eigenvalues. For a base point, fix module rates and let the hub rate d_h tend to 0. One eigenvalue tends to 0 and the others to the Hurwitz spectra of the scaled modules. The determinant comparison of Theorem 5.2 gives the small real eigenvalue $d_h(b + \sum_j G_j)(1 + o(1)) < 0$. Connectedness of the positive cone and continuity of eigenvalues conclude. $\square$

Theorem 5.2 is the case of one inner module (the inner block C) together with scalar modules $M = (-1)$, $w = 1$, $c = r_j$, whose value sets are the semicircles $\{r_j/(1 + it)\}$. Because the inner block of a D-stable matrix need not be D-stable, Theorem 5.2 treats singular inner pencils separately.

Corollary 9.4 (Negative-imaginary modules). *If every V_j lies in the closed lower half-plane, then the hub system is D-stable if and only if $-b - \sum_j G_j > 0$. This holds in particular for reciprocal relaxation modules.*

Proof. The ray has positive imaginary part. For reciprocal modules, $\text{Im} \sum_\ell g_\ell / (1 + i/\lambda_\ell) = -\sum_\ell g_\ell \lambda_\ell^{-1} / (1 + \lambda_\ell^{-2}) \leq 0$. $\square$

A stable scalar transfer function with $\text{Im} H(i\omega) \leq 0$ for $\omega > 0$ is negative imaginary. Corollary 9.4 is thus a D-stability counterpart of the theorem of Lanzon and Petersen [27, 34], in which a suitable DC gain condition decides stability of a negative-imaginary interconnection. The granularity phenomenon needs the opposite ingredient: a module, here the core's inner block, whose value set reaches the upper half-plane at the relevant real part. That happens exactly where $\Theta > 0$. For $B(6)$ this is precisely the open static window. Inside it, B_X has strict growth and $\det(-B_X) > 0$. It also has a Hurwitz scaling: with the port rate $v \rightarrow 0$ the third Hurwitz determinant tends to $F_0 > 0$ (proof of Proposition 8.1). So along a path of scalings from a stable one some eigenvalue crosses the imaginary axis away from zero; singular inner pencils carry no contacts (proof of Theorem 5.2). Outside its closure, B_X is D-stable.

10 Binding sites, decoys and buffers

Dictionary. Consider a linearized system $\dot{x} = DJx$ in which J is the flux Jacobian at a steady state. D is the positive diagonal matrix of inverse *capacities*: compartment volumes, buffering capacities or turnover scales, which are unknown. Let the port species bind to site classes $j = 1, \dots, m$. Class j has linearized binding coefficient k_j (on-rate times free-site concentration), unbinding rate u_j and degradation $\delta_j \geq 0$ of the bound complex y_j :

$$\dot{x}_p \ni -k_j x_p + u_j y_j, \quad \dot{y}_j \propto k_j x_p - (u_j + \delta_j) y_j.$$

In the notation of Section 2.3, $a_j = u_j$, $c_j = k_j$ and $h_j = u_j + \delta_j$. The normalized model has core $B = J_{\text{core}} - (\sum_j k_j) e_p e_p^T$ and loads

$$r_j = \frac{k_j u_j}{u_j + \delta_j},$$

with free attachment rates $(u_j + \delta_j)$ times the inverse capacity of y_j . A slow site class, whose complexes are effectively frozen, acts as a pure loss k_j on the port. A fast class equilibrates instantly, and its net effect on the port diagonal is $-k_j \delta_j / (u_j + \delta_j)$. The static loading interval therefore runs from the loss-only core B (all classes slow) to the fast-site Schur complement B_G (all classes fast). With no degradation the two ends are $J_{\text{core}} - \sum_j k_j e_p e_p^T$ and J_{core} itself.

What the theorems predict. Suppose that both ends of the interval are D-stable, so that neither a slow nor a fast binding environment destabilizes the core. If the interior of the interval contains static strict growth, then Theorem 4.3 and Proposition 6.2 predict that *heterogeneous* site populations can destabilize the core, while a single homogeneous class may not. The criterion is combinatorial: some grouping of the site classes into “fast” and “slow” must place the effective static damping of the port inside the unstable window. Only dimensionless comparisons enter: the loads r_j and their subset sums against the window of the static ray, all in the units of the core’s entries. Merging classes never hurts (Corollary 6.3). Bounding the ratio of kinetic time scales raises the threshold when $\det B_\tau \neq 0$ (Proposition 4.5).

The four-state example as a binding problem. Take $J_{\text{core}} = B(4)$ and site classes of total binding coefficient 2 with $\delta = 0$. The loss-only core is $B(6)$ and the fast limit is $B(4)$; both are D-stable. By Corollary 8.4, the system is D-stable for all capacities and unbinding rates if and only if one site class carries at least $6 - h_- \approx 1.8787$ of the total binding coefficient. In that case the remaining classes, whatever their kinetics, cannot move the port into the window. With two equal classes the linearization has a growing oscillatory mode. In the explicit witness the effective rates of the two classes differ by a factor of about 10^3 ; this ratio can be carried by the unbinding rates, by the capacities of the complexes, or by both. This is an exact linear interface example, not a calibrated biochemical model.

Scope in applications. Buffer- and decoy-induced oscillations, and their suppression, are known in specific nonlinear models [14, 16, 23, 12]. So is the loading effect of downstream binding [11, 20]. Adding sites to a nonlinear model generally changes its steady state and hence the core Jacobian, whereas our comparison holds the linearized core fixed. In particular, a fixed three-state core, such as a Goodwin-type loop, cannot exhibit the fixed-total granularity effect (Proposition 7.1). The theorems quantify over *all* positive rates: they are worst-case, time-scale-robust statements. In a specific kinetic model some rates may be linked, and the instability conclusions are then existence statements over a larger parameter set than the model allows. The conclusions concern linear spectral growth; no statement about Hopf bifurcations or limit cycles is made.

11 Verification and scope

Formal verification. Theorem 4.3 is verified end to end in Lean 4 [10] with Mathlib [31], including $0 \leq \tau \leq -B_{pp}$ and the complete equality case. The exported theorem is `granularity_threshold` in the namespace `DStabilityCharacterization.Granularity`, in the file `proofs/DStabilityCharacterization/Threshold.lean`. It is stated for arbitrary finite index types of core and attachments, with literal matrix eigenpairs under every positive diagonal scaling. D-stability is strict: every eigenvalue of every scaling has negative real part. The fineness ε in part (iii) is quantified before the attachment type and the partition. The proof formalizes Lemma 3.1, Theorem 3.2, Lemma 4.1, Lemma 4.2(a) with a uniform tolerance, and Lemma 2.1. The spread bound of Lemma 4.2(b) is not part of the formal statement. It assumes no spectral-continuation, coverage or realization premise. The strict build (Lean v4.30.0, warnings as errors) compiled with 19 project dependencies. The theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The verified source has SHA-256 2e515988238c2dcb4d8b0143f1aa510ca90c7d1bc90421633a24014a70055520.

Exact computation. Lemma 8.2 is an exact computer-assisted certificate. The script `check_paper.py` accompanies the source and replays it in rational and algebraic arithmetic. The same script checks every printed number: the identities of Lemmas 3.1 and 4.2, the three-state identities of Section 7, the coefficients and Hurwitz decomposition of Proposition 8.1, the constants $h_{\pm}$, τ and $6 - h_{-}$, the verdicts of Table 1, the witness polynomial and its Routh signs, and the refinement identity. It also checks the Lean receipt and its hashes. A few floating-point sanity checks are labelled as such.

Conventional proofs. All other results carry complete conventional proofs here and are not claimed as formally verified: Lemma 4.2(b), Proposition 4.5, Sections 5, 6 and 7, Theorem 8.3, Corollary 8.4, and Section 9. Proposition 8.1 is conventional reasoning on top of polynomial identities (the coefficients c_i and the decomposition $\Delta_3 = \sum_j F_j v^j$) that are checked by exact computation, as is the witness for $(1, 1)$.

Scope. The threshold τ and the envelope Θ are reductions, not algorithms. Deciding D-stability of B_X in general, and computing Θ for large cores, remain as hard as the underlying static problem. The coordinate structure of the interface is essential: the absorbed lag $\lambda \alpha e e^T$ is a legitimate change of the port's rate only because $e e^T$ is diagonal. For a non-coordinate interface uv^T the reduction fails.

12 Open problems

1. *Computing the core envelope.* Characterize Θ for small inner blocks, and decide when it is attained at equal inner rates, as it empirically is for $B(6)$. Sum-of-squares or Pólya certificates for the disk conditions of Corollary 5.3 give sufficient tests, and scaled small-gain relaxations give conservative bounds. An exact efficient method is not expected in general.
2. *When is the subset-sum test exact?* Characterize the cores for which D-stable corners imply D-stable edges, as in Lemma 8.2.
3. *Non-coordinate interfaces.* Find the smallest example in which phase, rather than static load, sets the threshold.
4. *The time-scale-limited threshold.* Compute τ_{κ} of Proposition 4.5, and its asymptotics as $\kappa \rightarrow \infty$, for the family of Section 8. Relatedly, determine the minimal spread of an unstable scaling as a partition approaches the boundary of the window.

5. *Random heterogeneity*. For random loads and log-uniform time scales, find the probability of instability as the number of pieces grows.
6. *Networks of modules*. Extend the hub criterion to trees of modules glued at single coordinates, via a recursion of planar value sets.
7. *Nonlinear onset*. Decide whether the crossing of the window by a subset sum produces a supercritical Hopf bifurcation in a realized kinetic model.

Data and code availability

The Lean sources, the verification receipt, `check_paper.py` and `make_figures.py` are part of the accompanying repository. The manifest `ARTIFACT_MANIFEST.json` lists their SHA-256 hashes.

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